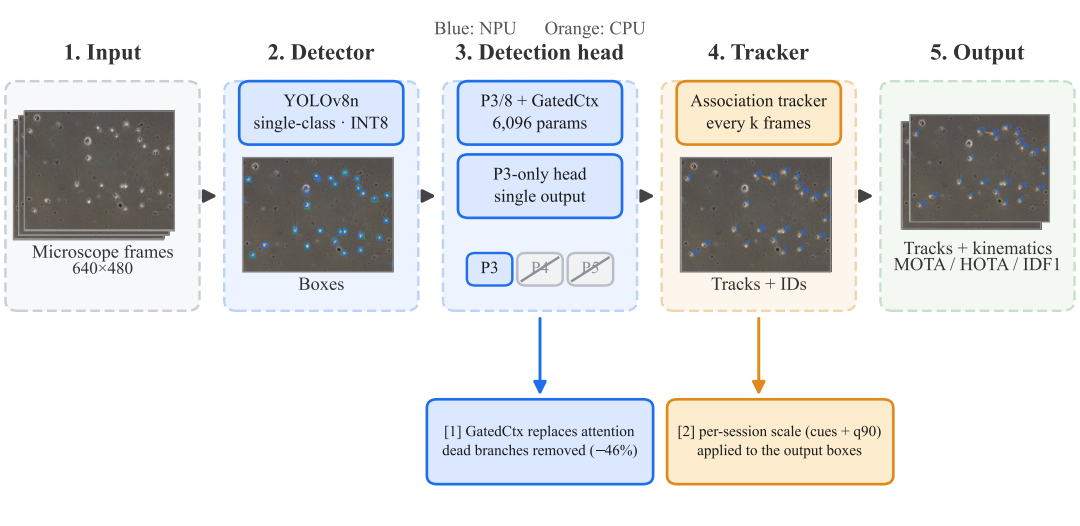

Blue: NPU Orange: CPU
1. Input
2. Detector
3. Detection head
4. Tracker
5. Output
Microscope frames
640×480
YOLOv8n
single-class · INT8
Boxes
P3/8 + GatedCtx
6,096 params
P3-only head
single output
P3
P4
P5
Association tracker
every k frames
Tracks + IDs
Tracks + kinematics
MOTA / HOTA / IDF1
[1] GatedCtx replaces attention
dead branches removed (−46%)
[2] per-session scale (cues + q90)
applied to the output boxes